DATA MINING AND BUSINESS INTELLIGENCE · CA2 MINI PROJECT · 10 MARKS

BlackSpot Miner: Mining Road Collision Data to Locate Severe-Crash Hotspots and Prioritise Road Safety Action

Implementation of the CA1 case study with DBSCAN, Apriori, K-Modes, Naive Bayes, Decision Tree, Random Forest and AdaBoost

Group D15A_9	Domain	SDG goal	Dataset	Code
Jaiprakash Talreja (54) Adarsh Singh (52) Vidushi Singh (53) Piyush Shukla (49)	Transportation and road safety	SDG 3 (target 3.6, road deaths and injuries); SDG 11 (sustainable cities)	DfT STATS19, 513,801 collisions, 2021 to 2025	GitHub repository, link in section 10

804
street-scale KSI hotspots found

107 of 109
rules held on unseen 2025 data

0.764
ROC-AUC for next-year site risk, against 0.680 for past counts

34.3%
of 2025 KSI collisions in the top 10,000 ranked cells

1. Goal carried over from CA1

CA1 framed the problem: a road authority has a limited safety budget and must decide where to act. This project builds that analysis on real data and answers three questions. **Where** do collisions that kill or seriously injure (KSI) concentrate? **Why** are some collisions more severe? **What next:** which sites should be treated first? Every result is tested on 2025, a year never used for mining or training.

2. Dataset and preprocessing

The Department for Transport publishes every police-reported injury collision in Great Britain [1]. We used the latest five years: 513,801 collisions, 937,265 vehicles and 652,821 casualties, including 8,033 people killed and 129,011 seriously injured.

Step	Result
Records removed (no location or speed limit)	56, leaving 513,745
Tables joined on `collision_index`	Vehicle and casualty detail added as flags: motorcycle involved, pedestrian hit, left the road, overturned and others
Leakage removed	`number_of_casualties`, police attendance and enhanced severity dropped: they are known only after the outcome
Unusable fields removed	`special_conditions_at_site` is 86.5% missing in 2025; `*_historic` fields are mostly empty in recent years
Codes mapped to labels	All from the official 2025 data guide. Unknown values kept as an explicit category

Consistent severity label	309,041 injury-based records for rules and severity models; DfT adjusted KSI for hotspots and location ranking
Time-based split	Severity: train 221,193 (2021 to 2024), test 87,848 (2025)

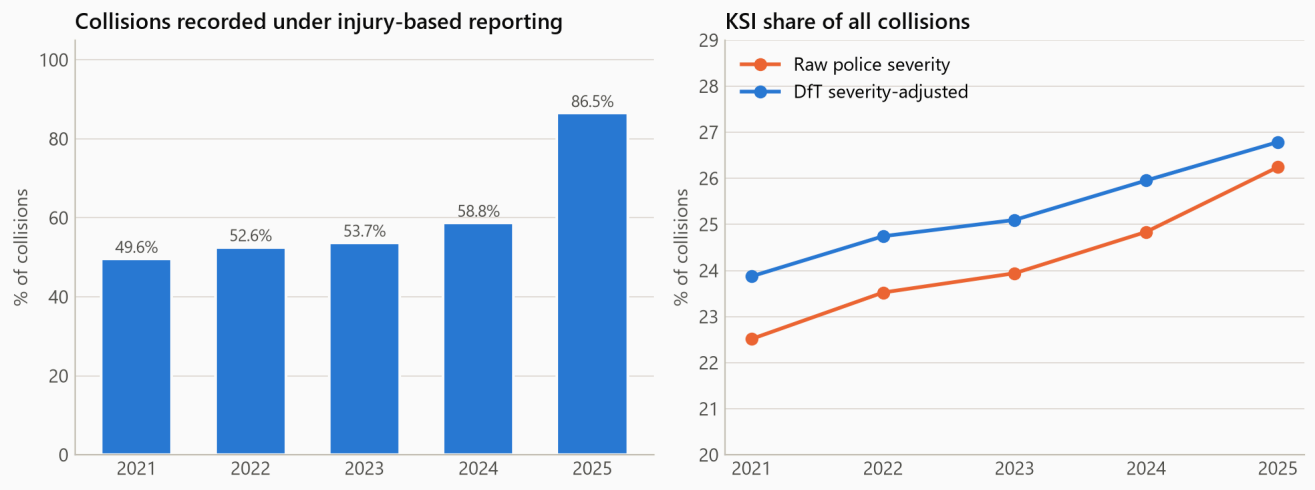


Figure 1. The injury-based share of collisions rose from 49.6% to 86.5%, and 15 of 44 force codes changed system. Without adjustment, raw KSI share overstates the rise.

3. Exploratory analysis

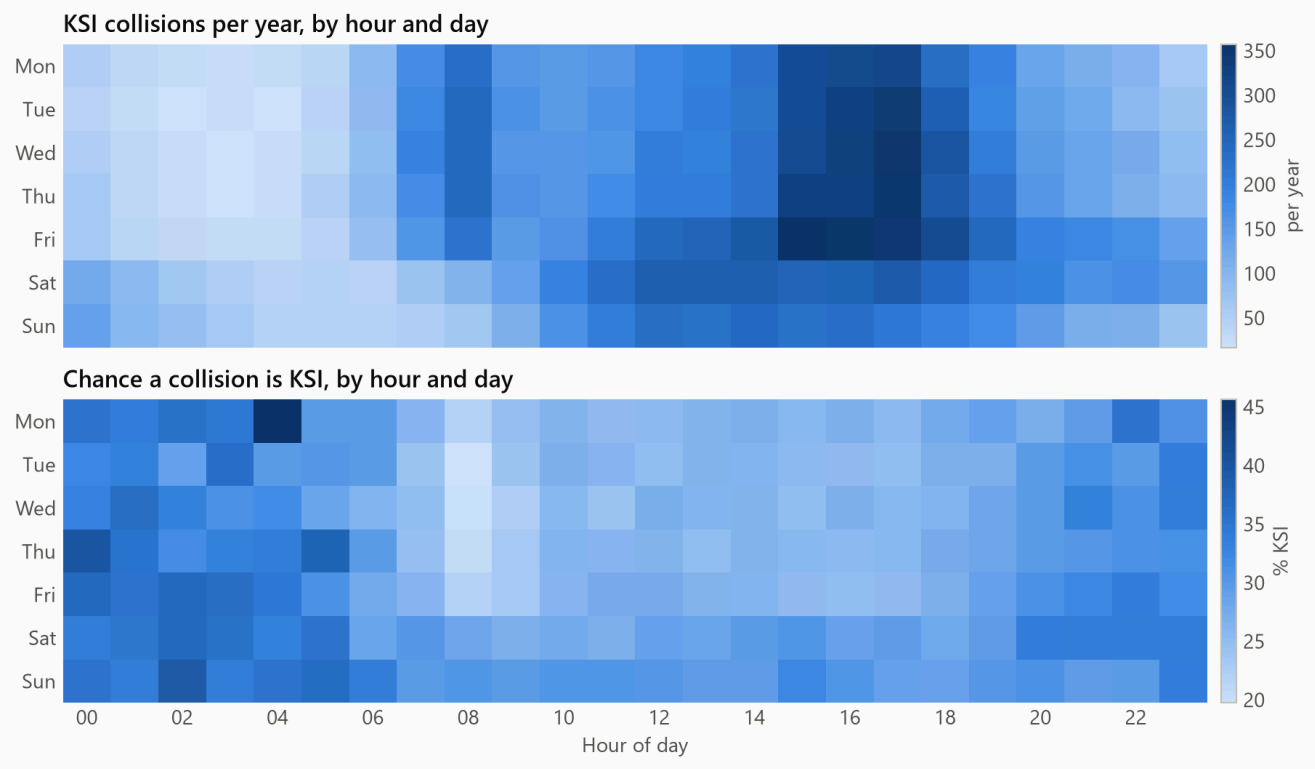


Figure 2. The number of KSI collisions peaks on weekday afternoons (Fri 15:00, about 357 a year). The chance of KSI peaks in the early hours, up to 45.6% (Mon 04:00) against a low of 19.8%. Volume and severity are different problems.

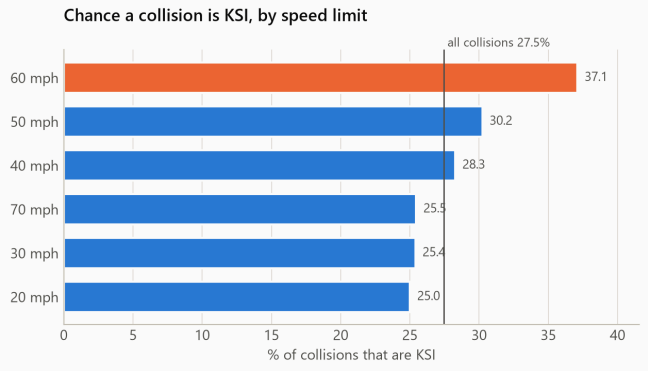


Figure 3. Chance of KSI by speed limit. 60 mph roads are worst at 37.1%; motorways at 70 mph are 25.5%.

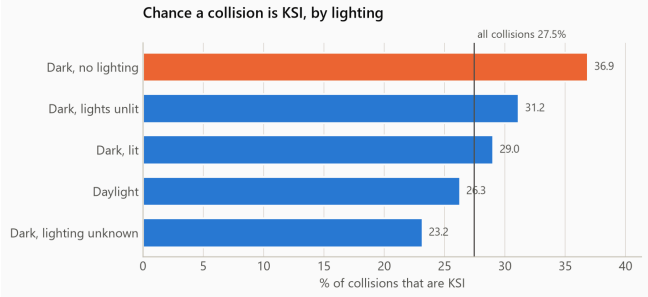


Figure 4. Darkness with no lighting reaches 36.9%, against 26.3% in daylight.

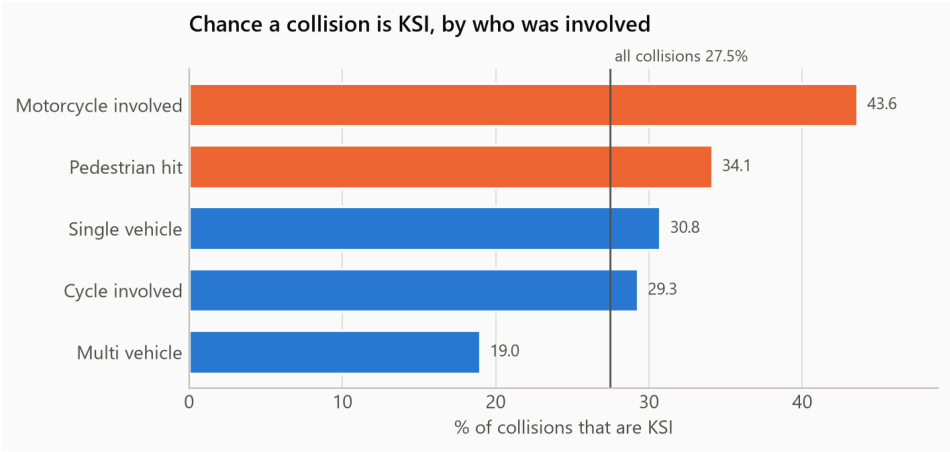


Figure 5. Who was involved matters most: 43.6% for motorcycles and 34.1% for pedestrians, against 19.0% for multi-vehicle collisions.

4. Where: hotspot discovery with DBSCAN (Module IV)

DBSCAN ran on the 124,459 KSI collisions, using grid coordinates in metres. Settings were chosen by test, not by habit. We ran 16 combinations and kept those whose largest cluster stayed under 1 km, so a hotspot stays a place an engineer can visit. The largest number of hotspots under that limit came from `eps = 100 m` and `min_samples = 6`.

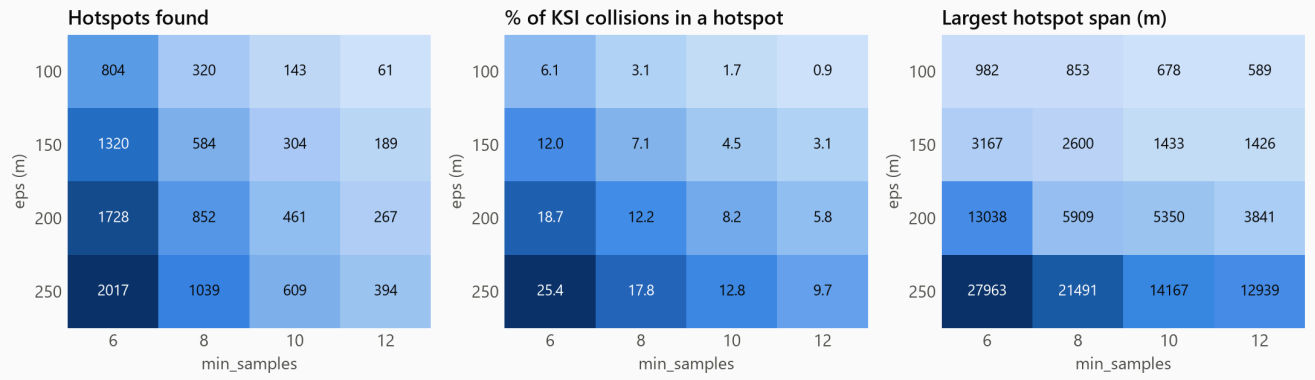


Figure 6. Settings test. At 250 m with low min_samples the largest cluster runs to tens of kilometres because hotspots chain together across London.

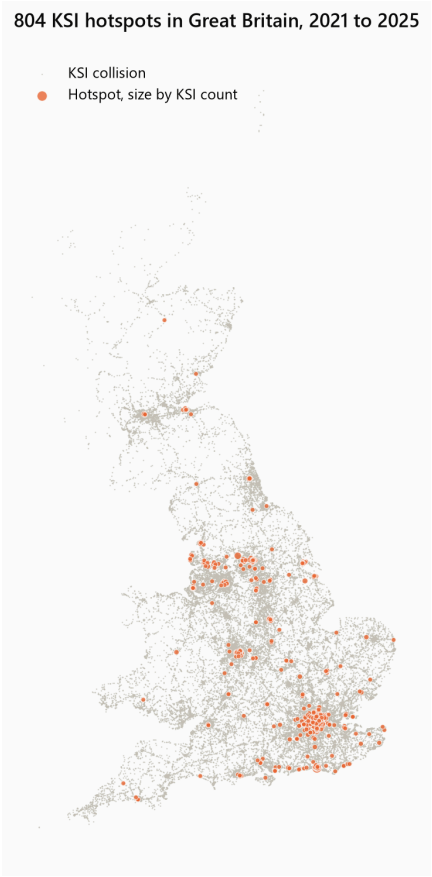


Figure 7. 804 hotspots. Median span 173 m, largest 982 m.

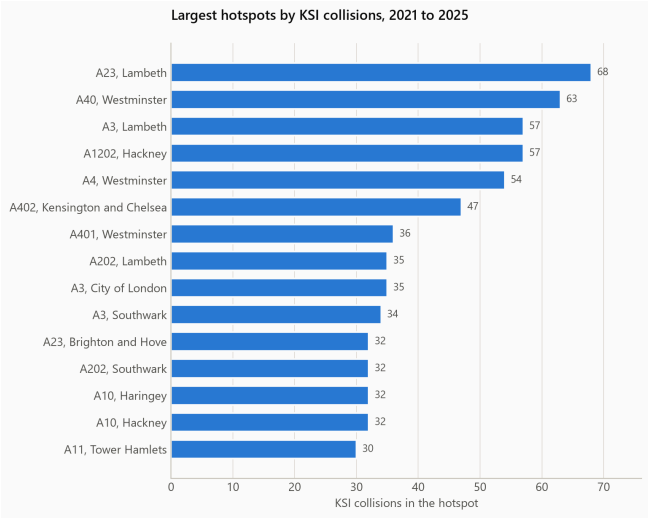


Figure 8. The largest hotspots are mostly on central London A roads.

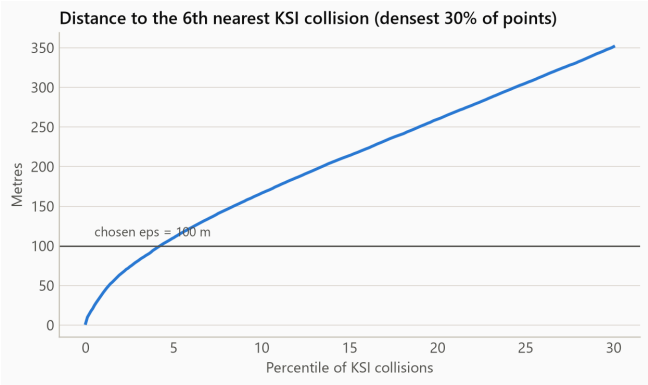


Figure 9. k-distance curve confirming the chosen eps sits in the dense part of the data.

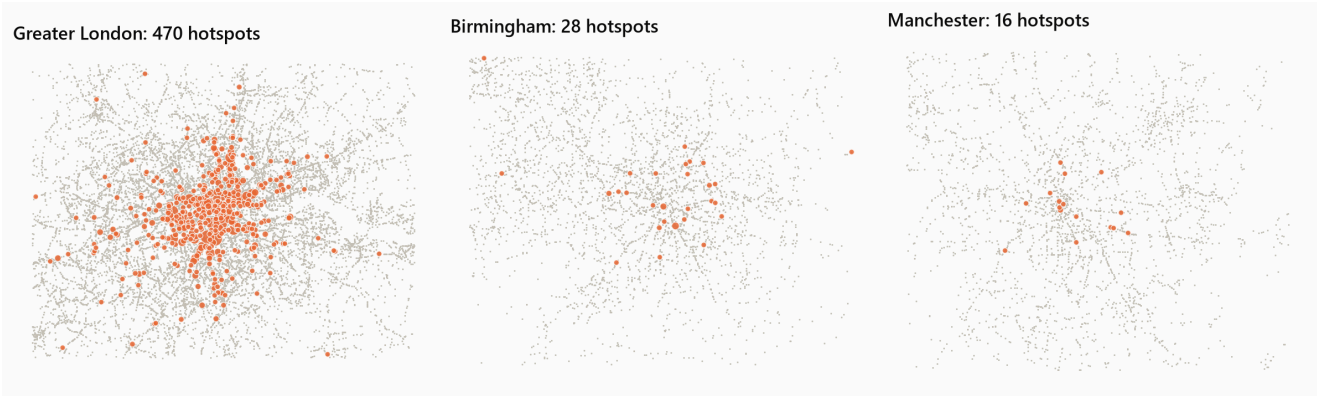


Figure 10. City views. Grey points are KSI collisions; orange circles are hotspots.

Rank	Road	Authority	KSI	Fatal	Span	Main pattern
1	A23	Lambeth	68	5	885 m	Cycle
2	A40	Westminster	63	1	909 m	Cycle
3	A3	Lambeth	57	2	831 m	Cycle
4	A1202	Hackney	57	1	982 m	Cycle
5	A4	Westminster	54	1	832 m	Cycle

Do hotspots persist? We rediscovered hotspots on 2021 to 2024 only, then counted 2025 KSI collisions within 100 m. They reached 3.16% of 2025 KSI collisions, against 2.37% for the same number of random past KSI locations. Hotspot points overlap, so they cover only 35.1 km² against 126.0 km². Per square kilometre they collected 4.79 times as many 2025 KSI collisions (23.97 against 5.0).

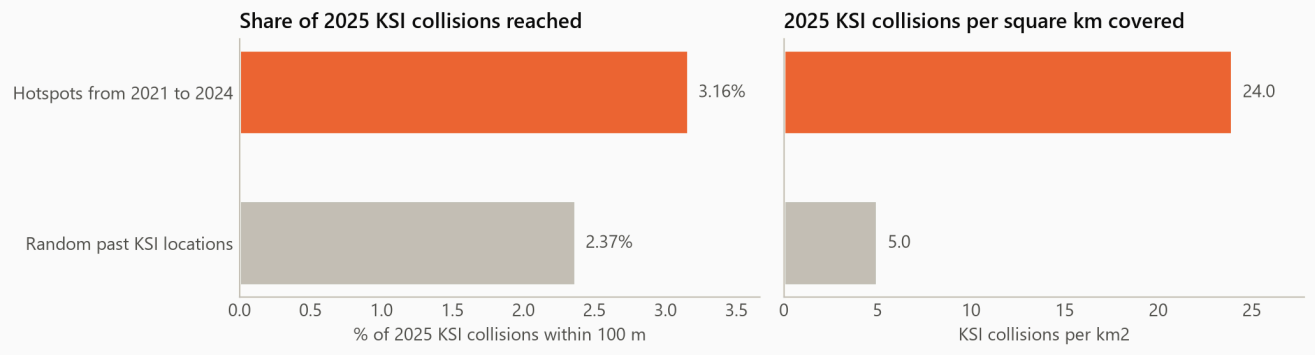


Figure 11. Hotspots reach modestly more 2025 KSI collisions in total, but far more per square kilometre.

Reading this honestly. 93.89% of KSI collisions fall outside any hotspot. Treating hotspots alone cannot solve the problem. Most severe crashes are scattered, so site work needs to sit beside route and area programmes.

5. Why: association rules with Apriori (Module V)

Apriori ran on 221,193 injury-based collisions from 2021 to 2024, with 63 items, minimum support 1% and at most four items per itemset. It took 19.9 seconds and produced 20,684 frequent itemsets and 1,657 rules ending in KSI. Requiring lift of at least 1.3 and confidence of at least 35% left 579. Dropping rules whose shorter version already reached the same confidence left 109. The KSI base rate was 27.64% in the mined period and 27.0% in 2025.

Strongest rule per theme (antecedent → KSI)	Support	Conf.	Lift	Conf. 2025	Lift 2025
Left the road + Motorcycle involved	1.35%	66.7%	2.41	59.1%	2.19
Left the road + Speed: 60 mph + Surface: Dry	1.68%	46.0%	1.66	44.2%	1.64
Left the road + Overturned + Surface: Dry	1.11%	44.2%	1.60	40.8%	1.51
Light: Dark, lit + Pedestrian hit	1.72%	42.7%	1.55	38.5%	1.43
Layout: Single carriageway + Left the road + Light: Dark, no lighting	1.06%	40.2%	1.45	38.8%	1.44
Hit tree or pole	1.25%	39.6%	1.43	38.3%	1.42
Area: Rural + Cycle involved	1.06%	39.3%	1.42	39.8%	1.47
Area: Rural + Driver 70 or over	1.97%	36.7%	1.33	37.6%	1.39

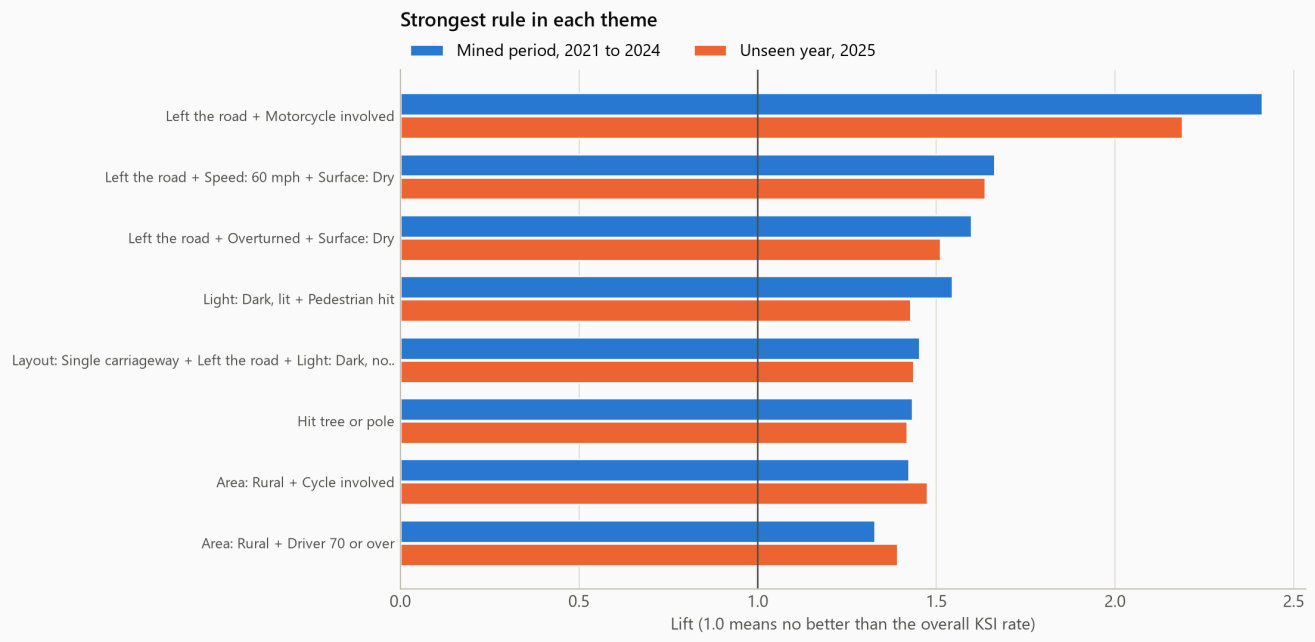


Figure 12. The same rules on the mined years and on unseen 2025. Lift barely moves.

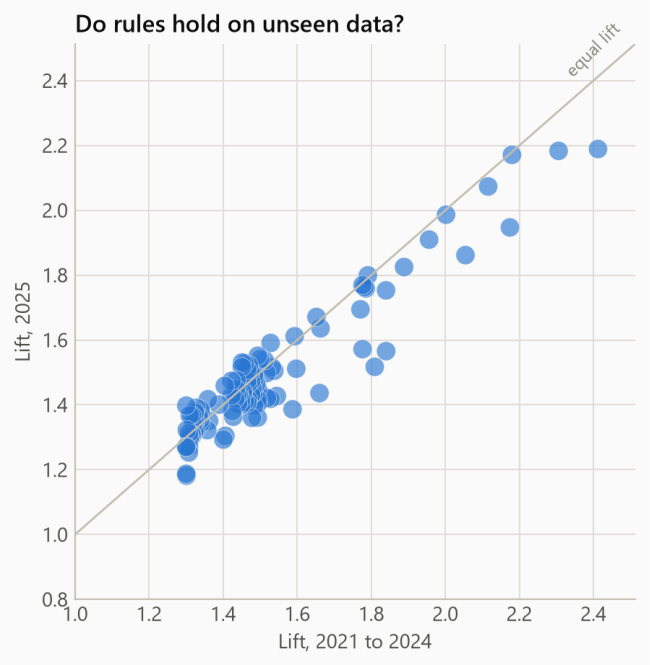


Figure 13. 107 of 109 rules keep lift of at least 1.2 on 2025. Correlation of lift between periods is 0.946; median shrinkage is 0.9%.

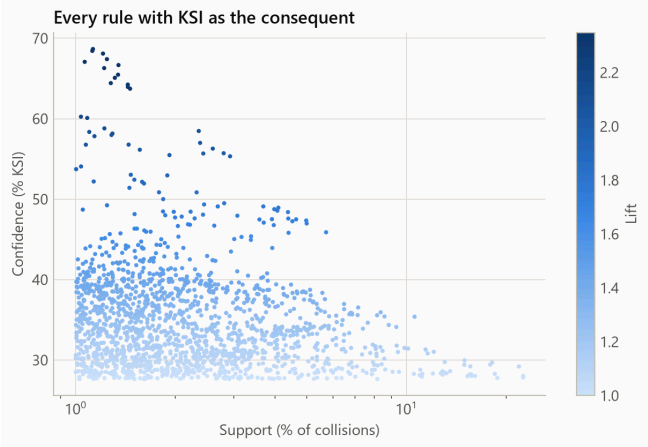


Figure 14. All KSI rules. High confidence usually comes with low support: the strongest patterns are specific.

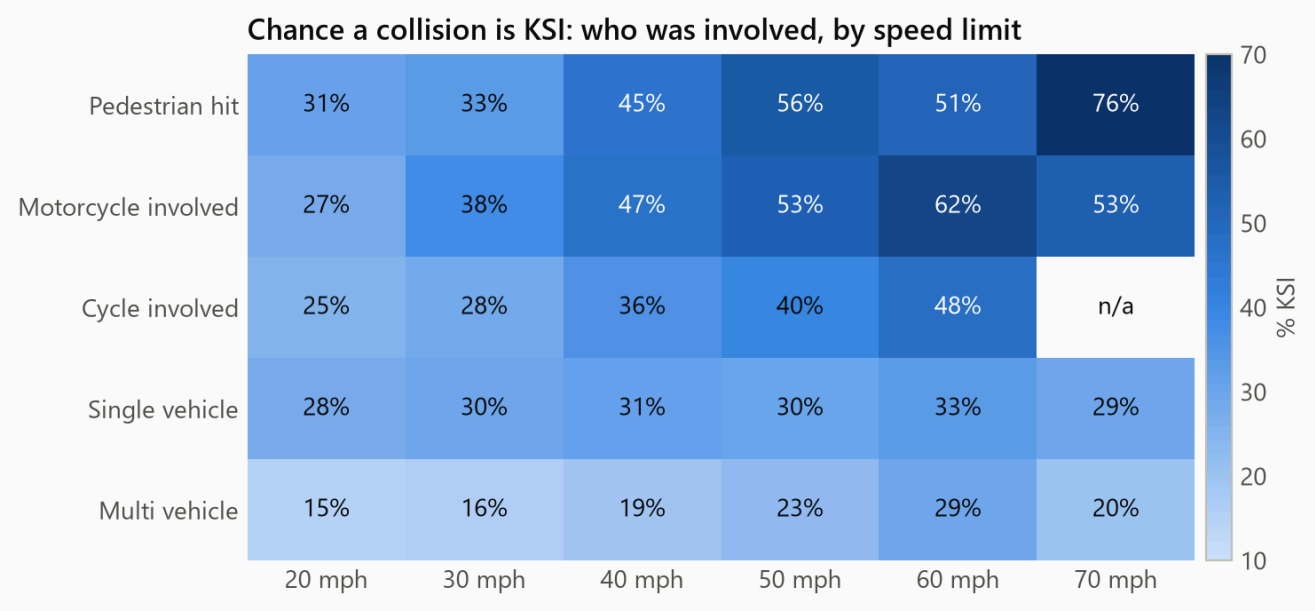


Figure 15. Interaction between road user and speed limit. A pedestrian hit is KSI 33% of the time at 30 mph, 46% at 40 mph and 51% at 60 mph. Multi-vehicle collisions rise much less. Cells under 200 collisions show n/a.

Testing the reference paper's claim

Kumar and Toshniwal [2] argue that segmenting with K-Modes first reveals rules that the entire data set hides. We repeated this with their k = 6 on our data. The cost curve in Figure 16 has no sharp elbow, so we did not claim an optimal k. We then added a test they did not run: whether those rules hold on 2025.

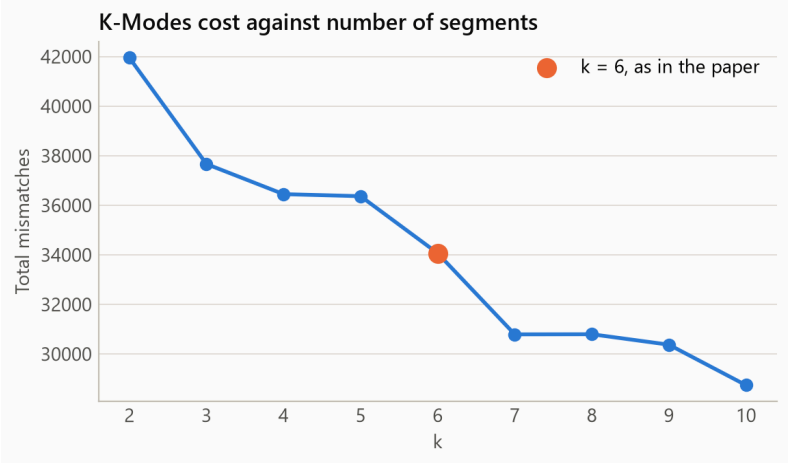


Figure 16. K-Modes cost on a 20,000 row sample.

Seg.	Most common profile	Share	KSI rate
S1	Multi vehicle, Unclassified, 30 mph, Daylight, Urban	31.5%	23.9%
S2	Multi vehicle, A road, 60 mph, Daylight, Rural	28.5%	29.7%
S3	Single vehicle, A road, 60 mph, Dark, no lighting, Rural	4.6%	38.6%
S4	Motorcycle involved, A road, 30 mph, Daylight, Urban	15.2%	31.3%
S5	Pedestrian hit, Unclassified, 30 mph, Daylight, Urban	12.3%	30.8%
S6	Multi vehicle, Unclassified, 30 mph, Daylight, Urban	7.9%	16.7%

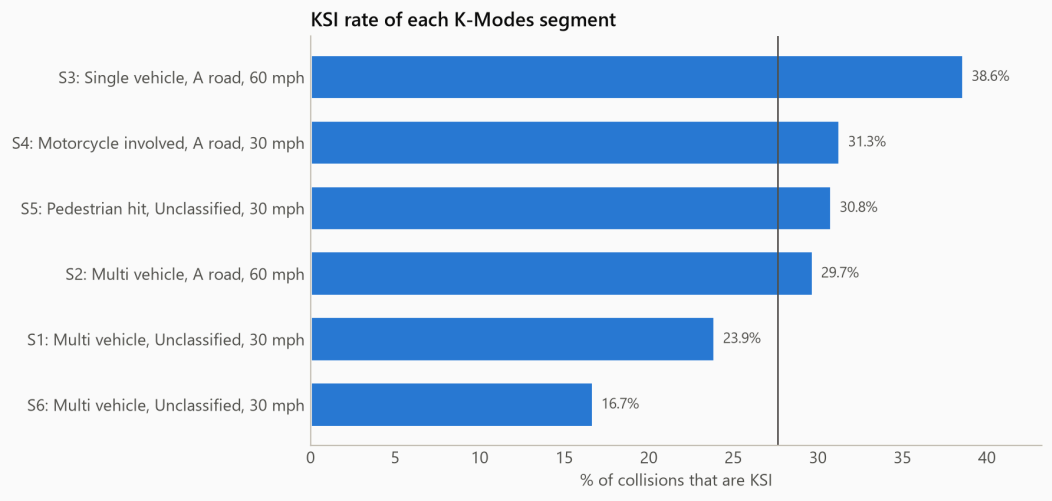


Figure 17. Segment KSI rates range from 16.7% to 38.6%. Segments split collisions into genuinely different risk groups.

Seg.	Rules	Hidden	Hold 2025	Top hidden rule	Lift
S1	28	3	3	Road: Unclassified + Single vehicle	1.37
S2	78	16	16	Day: Weekend + Layout: Single carriageway	1.36
S3	17	5	5	Light: Dark, no lighting + Pedestrian hit	1.78
S4	69	17	14	Road: Unclassified + Weather: Fine	1.39
S5	13	9	9	Light: Dark, lit + Surface: Wet	1.26
S6	13	10	8	Area: Urban + Left the road	1.65

Inside the segments we found 218 rules with lift of at least 1.2, of which 60 did not appear in the whole-data rules. 55 of those held on 2025 with lift of at least 1.1. **This supports the paper's claim on a much larger data set from a different country.**

6. Severity classification (Module III)

Four classifiers were tuned on 2021 to 2023 against 2024, refit on 2021 to 2024 and tested once on 87,848 collisions from 2025. Each decision threshold came from 2024 by maximising F1.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Random Forest	**56.9%**	**35.7%**	**74.4%**	**0.482**	**0.679**	**0.440**
AdaBoost	56.9%	35.4%	72.3%	0.475	0.670	0.426
Decision Tree	54.7%	34.5%	75.6%	0.474	0.665	0.422
Naive Bayes	54.4%	33.8%	71.7%	0.460	0.638	0.376

Random Forest scored best with PR-AUC 0.440 (95% bootstrap interval 0.4335 to 0.4456), against 0.27 by chance, and ROC-AUC 0.679 (0.6749 to 0.6822). The confusion matrices show the trade-off clearly.

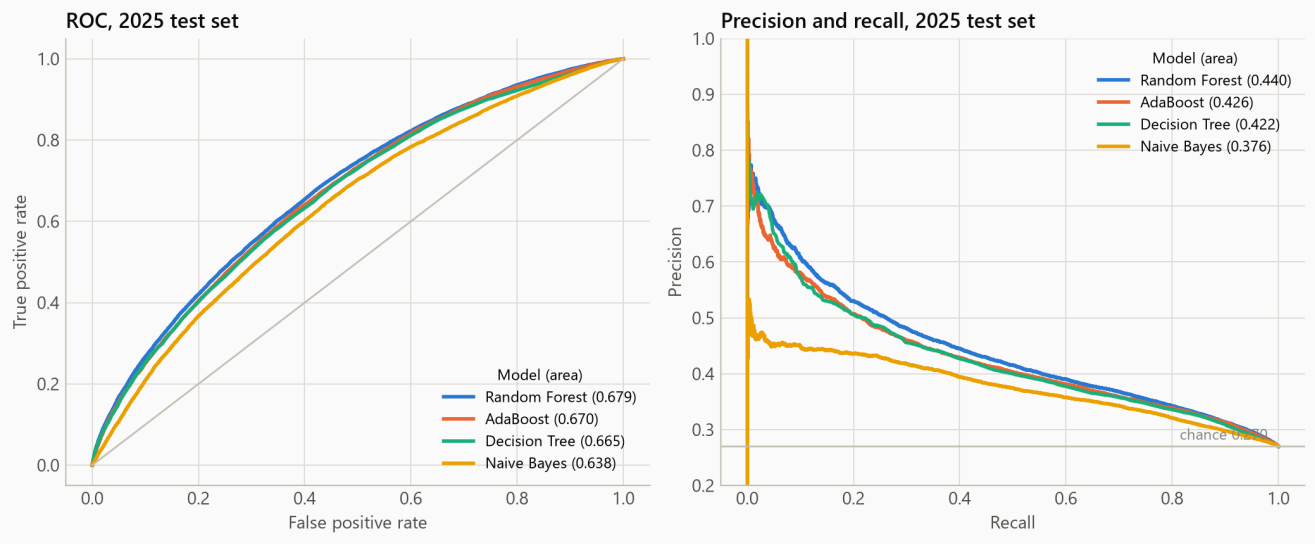


Figure 18. ROC and precision-recall curves on the 2025 test set.

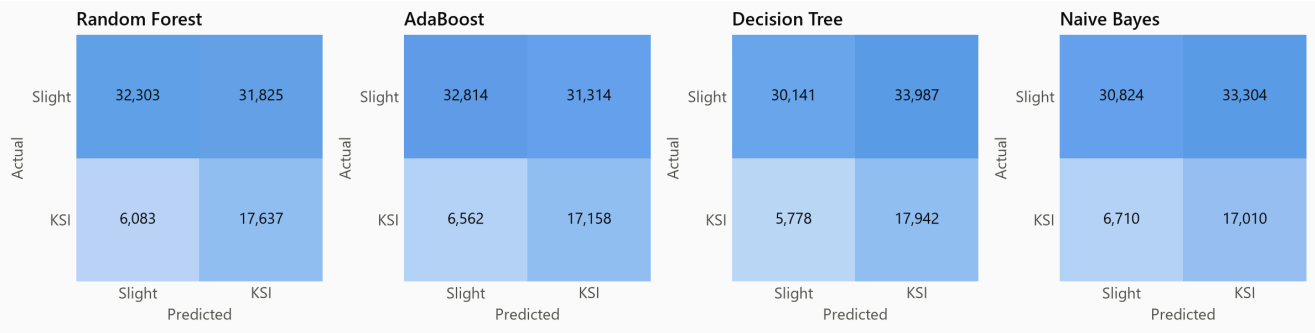


Figure 19. Confusion matrices at the 2024 thresholds.

Why accuracy is low and why that is correct. Predicting Slight for everything scores 73.0% accuracy and catches no KSI collision. The threshold here favours recall, so accuracy falls while 74.4% of KSI collisions are caught. At a calibrated 0.5 cut-off, accuracy rises to 73.9% but recall falls to 14.3%.

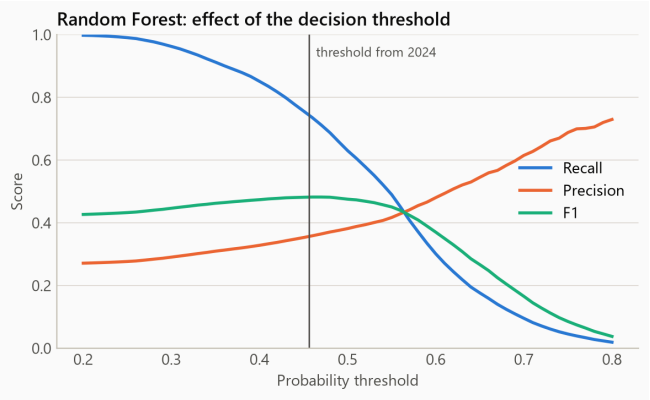


Figure 20. Moving the threshold trades recall against precision.

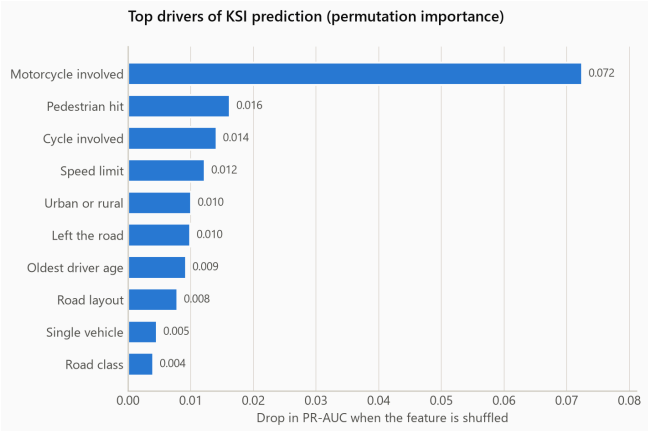


Figure 21. Permutation importance on 2025. Road user type leads.

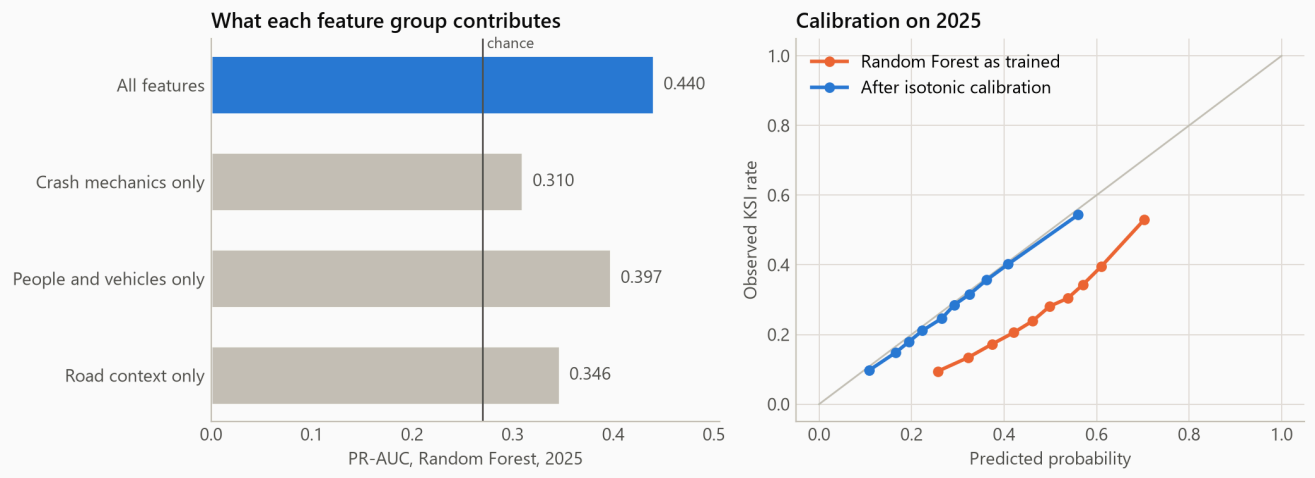


Figure 22. Left: road context alone gives PR-AUC 0.346; adding people and crash mechanics reaches 0.440. Right: balanced class weights inflate probabilities, and isotonic calibration on 2024 corrects them (Brier score 0.2238 to 0.181).

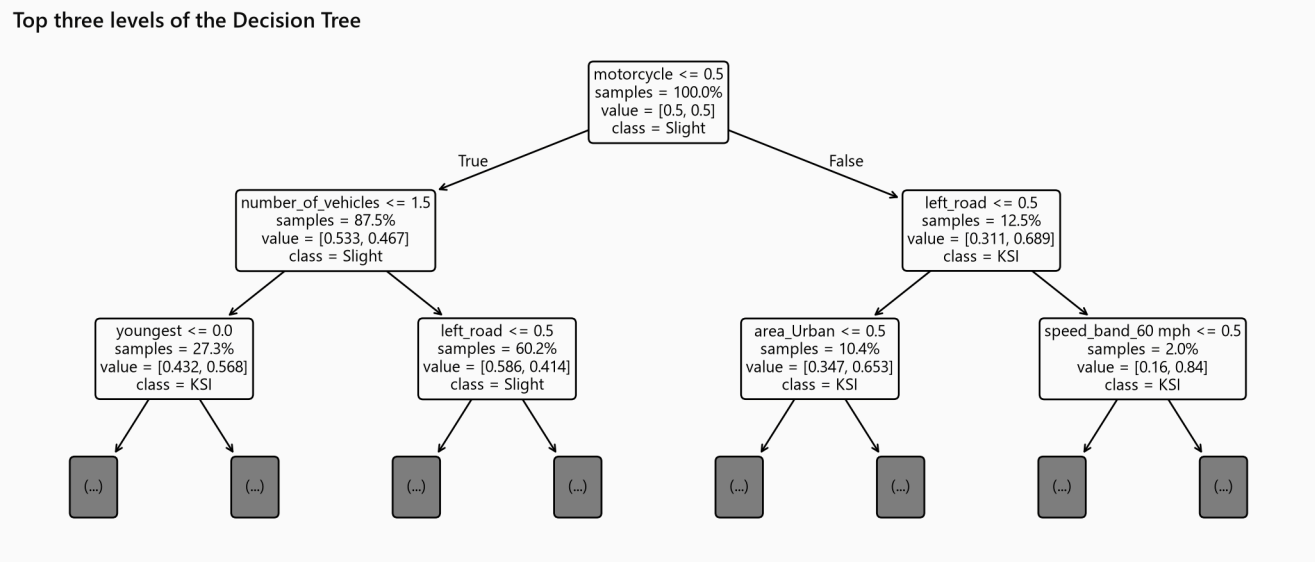


Figure 23. Top of the Decision Tree: the readable version of the same pattern.

What this tells us. Performance is moderate, and that is a finding in itself. Severity depends on things STATS19 does not hold, such as impact speed, seat belt use and the health of the person hit. A much higher score on data like this would be a reason to check for leaked columns or a random split. Road context alone scores PR-AUC 0.346 against 0.27 by chance, so a severity model is useful for explaining risk, not for predicting single crashes.

7. What next: next-year location risk (Module III)

A budget is spent on places, so this model predicts at 500 m cell level. Features from three past years predict the following year: 2021 to 2023 to predict 2024 for training, and 2022 to 2024 to predict 2025 for testing. Totals use DfT adjustment, so forces on different reporting systems compare fairly. 95,806 cells were scored, of which 16.42% recorded KSI in 2025. The baseline is a traditional black spot list: rank by past KSI count.

Ranking method	ROC-AUC	PR-AUC	KSI in top 5,000
Poisson boosting	**0.764**	**0.460**	**24.1%**
Random Forest	0.761	0.456	24.0%
AdaBoost	0.762	0.455	23.7%
Decision Tree	0.759	0.442	23.5%
Past KSI count (baseline)	0.680	0.381	22.0%

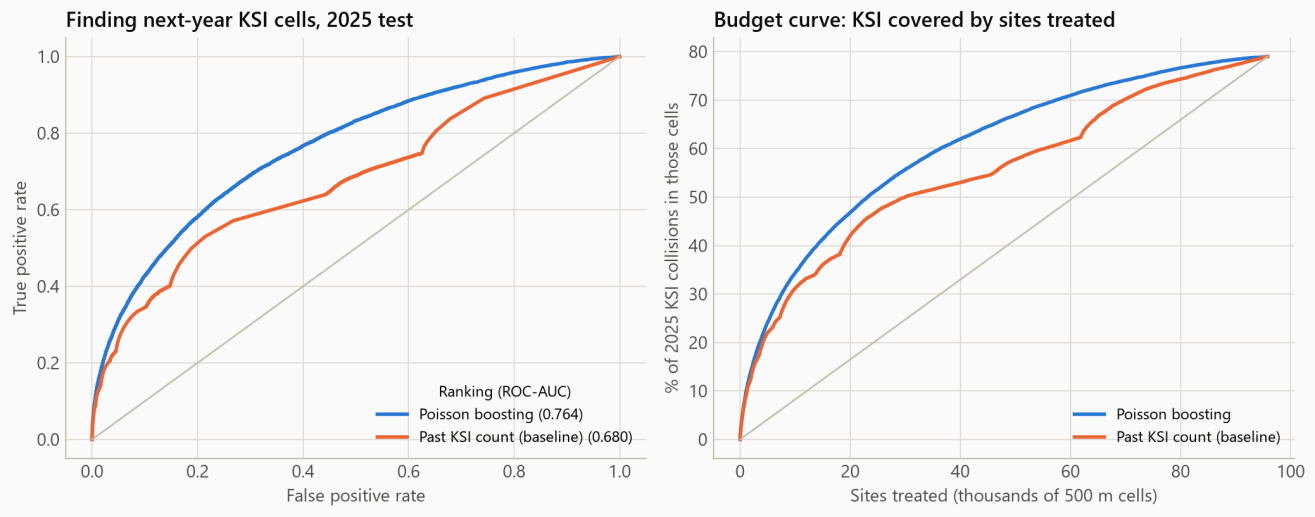


Figure 24. Poisson boosting against the past-count list. The grey line is what a random order of the same cells would cover.

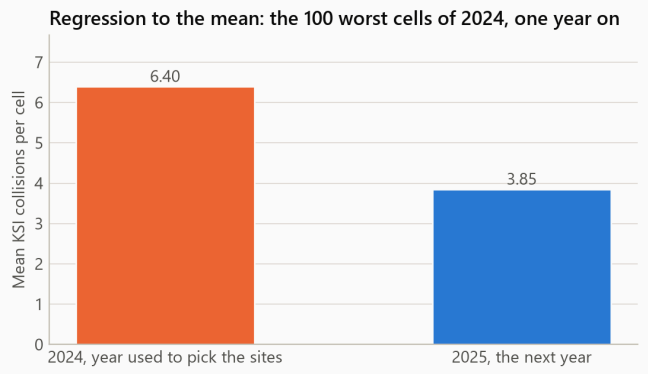


Figure 25. The 100 worst cells of 2024 fell 39.8% in 2025. The data records no treatment, and this is what regression to the mean predicts.

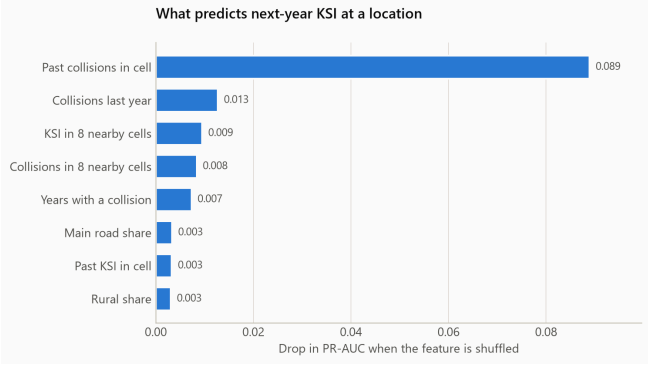


Figure 26. A site's collision history and its neighbours carry most of the signal.

Sites treated	Past-count list	Poisson boosting	Gain, points
100	1.8%	1.8%	-0.0
500	5.4%	5.8%	+0.4
1,000	8.6%	9.3%	+0.7
5,000	22.0%	24.1%	+2.1
10,000	31.2%	34.3%	+3.2

At 5,000 sites the model covers 2.13 more percentage points of 2025 KSI collisions than the past-count list (95% interval 1.78 to 2.53). The gain is small at the very top of the list and real further down. Past counts are

already a strong signal, and the model's value is in sites that a raw count misses. 78.98% of 2025 KSI collisions occurred in cells with a past collision; the rest are out of reach of any history-based ranking.

8. Insights

1. **Severe crashes are widely scattered.** 93.89% sit outside any hotspot, so hotspot work must be paired with route and area programmes.
2. **Vulnerable road users drive severity.** Motorcycles lead every ranking, in rules, importance and the tree. This matches the two-wheeler clusters in the Dehradun paper, despite very different traffic.
3. **Speed multiplies vulnerability.** The rise with speed limit is much steeper for pedestrians and motorcycles than for multi-vehicle collisions.
4. **Patterns are stable.** 98.2% of rules kept their strength on unseen data, so they can guide reviews.
5. **Worst-site lists flatter themselves.** A 39.8% fall in one year, with no treatment recorded in the data, means any before-and-after claim on such a list needs a control group.
6. **History predicts better than conditions.** Where a crash has happened is more informative than the conditions on the day.

9. Decision: how this achieves the goal

The model was refit on the latest window and scored on 2023 to 2025 to give a forward list for 2026. Each site receives a suggested review type taken from its own collision mix, compared with the national mix. 98.0% of the top 100 sites lie within 750 m of a DBSCAN hotspot, so the two methods agree on where the problem is.

Rank	Road	Authority	Collisions 2023-25	KSI	Suggested review
1	A3	City of London	73	23.0	Cycle separation and junction protection review
2	A201	City of London	71	28.0	Cycle separation and junction protection review
3	A201	Southwark	83	19.0	Cycle separation and junction protection review
4	A3	Southwark	77	19.0	Cycle separation and junction protection review
5	A401	Westminster	150	35.0	Pedestrian crossing and speed review
6	A40	Westminster	89	21.0	Cycle separation and junction protection review
7	A40	Westminster	150	29.0	Cycle separation and junction protection review
8	A107	Hackney	66	9.0	Cycle separation and junction protection review
9	A4	Westminster	67	19.0	Cycle separation and junction protection review
10	A302	Westminster	70	13.0	Cycle separation and junction protection review

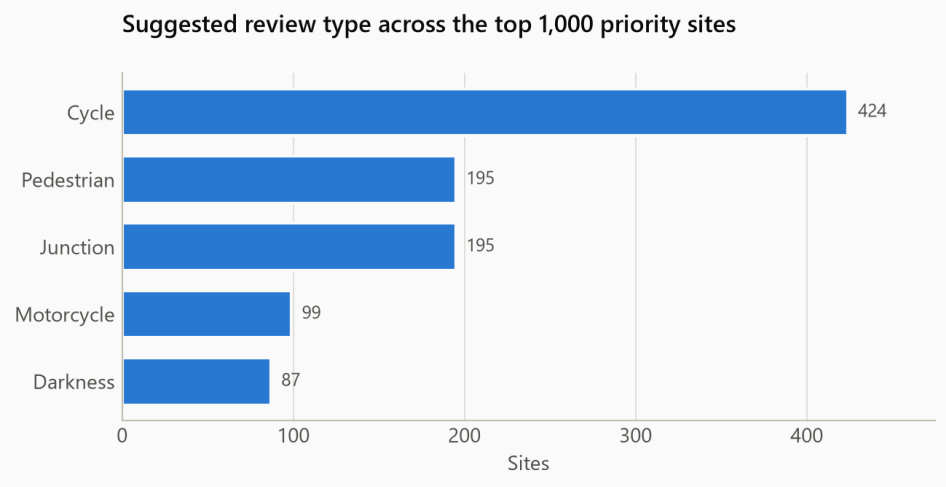


Figure 27. Suggested review types across the top 1,000 sites.

The authority's workflow is: open the interactive map (`blackspot_map.html`), take the shortlist that its budget allows from the budget table, read the site's dominant pattern, and send an engineer to verify it. The full list of 1,000 sites is in `priority_sites_2026.csv`.

SDG link. Target 3.6 addresses road deaths and injuries directly [3]. Target 11.2 on safe transport is listed as a related target. In India the same pipeline could run on eDAR records, which capture GPS-tagged accidents in all States and UTs [4].

Limits of this work

- STATS19 covers police-reported injury collisions only. Damage-only crashes and some unreported injuries are absent.
- There are no traffic volumes, so busy places rank high partly because they are busy. A rate per vehicle would need flow data.
- Rules and models show association, not cause, and no crash reduction is claimed for any review type.
- Review types come from a simple rule on collision mix. They are prompts for an engineer, not model predictions.
- Severity adjustment is a DfT statistical estimate. The location model inherits its uncertainty.
- Results describe Great Britain. The method transfers to India; the numbers do not.

10. Code and reproducibility

```
# hotspots: distance is in metres on the British National Grid
labels = DBSCAN(eps=EPS, min_samples=MS, algorithm="kd_tree", n_jobs=-1).fit_predict(xy)

# rules: mined on 2021-2024, then checked on 2025
freq = apriori(items_train, min_support=.01, use_colnames=True, max_len=4, low_memory=True)
rules = association_rules(freq, num_itemsets=len(items_train), metric="lift", min_threshold=1.0)
rules = rules[rules["consequents"] == frozenset({"KSI"})]

# next-year site risk: features from 2022-2024, target from 2025
X_te, y_te_cell = cell_frame([2022, 2023, 2024], 2025)
```

GitHub: https://github.com/Anexus5919/DMBI-CAs-VESIT_D15A_Gr-09-Roll.no.54-52-53-49

The notebook downloads the data from official URLs and runs end to end in about 22 minutes with fixed seeds. Every number in this report is read from `results/metrics.json`, which the notebook writes. Libraries: pandas 3.1.0, scikit-learn 1.9.1, mlxtend, kmodes, folium.

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